Recipe Rasoi: A Personalized Recipe Recommendation System for Pakistani Cuisine

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I. Introduction

Pakistani cuisine, known for its rich diversity and intricate flavors, often relies on specific ingredients and cooking methods. These dishes are passed down through generations, with each family adding its own variations to preserve the cultural legacy. However, many people face challenges in cooking authentic Pakistani food due to not always having access to all the required ingredients or wanting to experiment with new dishes. Moreover, there are people who live by themselves and want to cook Pakistani dishes but they don't have a proper resource to get such a recipe by giving the available ingredients. This way one may get deprived of having a proper home-cooked meal or have a lack of variety.

As a result, there is a clear need for a more flexible, personalized approach to cooking that allows individuals to create authentic dishes while adjusting for available ingredients and preferences. Hence, the objective of our project is to create a recipe recommendation system that provides user with a relevant Pakistani cuisine recipe according to the ingredients that the user has available.

II. Problem Definition

In today's fast-paced world, many individuals, particularly those living alone or in regions with limited access to traditional Pakistani ingredients, struggle to prepare authentic meals. The absence of a dynamic, personalized recipe recommendation model forces people to either settle for a bland meal which may be unfulfilling. Additionally, current static recipe platforms do not address the need for real-time recipe recommendation based on available ingredients or user preferences. Moreover, there is no such dataset present that has all the Pakistani recipes combined. This project addresses this gap by developing a recipe recommendation system that uses natural language processing and machine learning techniques to suggest personalized recipes tailored to the user's available ingredients.

III. Baseline Approach

For our project, we have chosen Support Vector Machine (SVM) model. A Support Vector Machine (SVM) is a powerful machine learning algorithm widely used for both linear and nonlinear classification, as well as regression and outlier detection tasks. The core idea behind SVM is to find the optimal hyperplane that separates the data points belonging to different classes (in our case, recipes) with the maximum margin.

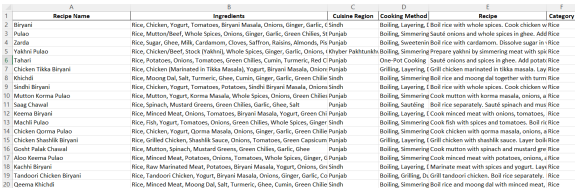
SVM is effective due to its focus on finding the maximum separating hyperplane between the different classes in the target feature. This is good for multi-class classification. The primary objective of the SVM algorithm is to identify the optimal hyperplane in an N-dimensional space that can effectively separate data points into different classes in the feature space. SVM model is good for our recipe recommendation model because of

the following points:

- **High-Dimensional Data Handling:** In a recipe recommendation model, each ingredient can be treated as a feature. A dataset with a wide variety of ingredients (like spices, proteins, and vegetables) can lead to high-dimensional data. Therefore, our dataset has many potential ingredients that can be put together. SVM solves this problem by managing this high-dimensional data. Each unique combination of ingredients is handled efficiently.
- **Non-Linear Relationships:** The relationship between ingredients and recipes is non-linear which means certain ingredients might make sense together and some recipes may have overlapping ingredients which makes the separation of classes as complex task. SVM handles this using RBF kernels to separate recipe classes easily.
- **Robustness with Complex Decision Boundaries:** Recipes often involve a combination of various ingredients, and predicting the correct recipe involves understanding these interactions.The use of kernels allows SVM to draw complex decision boundaries between different recipes which works better than simple models like logistic regression.
- **Ability to Handle Outliers and Ambiguity:** Recipes with similar ingredients but slight variations might belong to different classes. SVM focuses on the most important data points that are close to the dividing line between different classes (recipes). These key points help the model learn better, especially for tricky cases where the ingredients are similar but belong to different recipes. This makes the model good at handling small ingredient changes without getting confused.

IV. Data Description

Due to the lack of available datasets on Pakistani cuisine, we had to generate our own dataset. We generated this dataset using ChatGPT with the prompt that asked to give Pakistani Cuisine recipes. The dataset consists of a collection of Pakistani recipes, including information about ingredients, cooking methods, and cuisine regions.There are over 3,000 recipes, categorized by dish type, regional origin, and ingredients. Figure 1 shows some of the recipes in the excel file. Furthermore, Our data is divided into different categories and the division is shown through Figure 2.



	A	B	C	D	E	F
1	Recipe Name	Ingredients	Cuisine Region	Cooking Method	Recipe	Category
2	Biryani	Rice, Chicken, Yogurt, Tomatoes, Biryani Masala, Onions, Ginger, Garlic, C	Sindh	Boiling, Layering, I	Boil rice with whole spices. Cook chicken w	Rice
3	Pulao	Rice, Mutton/Beef, Whole Spices, Onions, Ginger, Garlic, Green Chilies, St	Punjab	Boiling, Simmering	Sauté onions and whole spices in ghee. Add	Rice
4	Zarda	Rice, Sugar, Ghee, Milk, Cardamom, Cloves, Saffron, Raisins, Almonds, Pis	Punjab	Boiling, Sweetenin	Boil rice with cardamom. Dissolve sugar in	Rice
5	Yakhni Pulao	Rice, Chicken/Beef, Stock (Yakhni), Whole Spices, Ginger, Garlic, Onions,	Khyber Pakhtunkh	Boiling, Simmering	Prepare yakhni by simmering meat with spi	Rice
6	Tahari	Rice, Potatoes, Onions, Tomatoes, Green Chilies, Cumin, Turmeric, Red C	Punjab	One-Pot Cooking	Sauté onions and spices in ghee. Add potat	Rice
7	Chicken Tikka Biryani	Rice, Chicken (Marinated in Tikka Masala), Yogurt, Biryani Masala, Onion	Punjab	Grilling, Layering, I	Grill chicken marinated in tikka masala. Lay	Rice
8	Khichdi	Rice, Moong Dal, Salt, Turmeric, Ghee, Cumin, Ginger, Garlic, Green Chilie	Sindh	Boiling, Simmering	Boil rice and moong dal together with turm	Rice
9	Sindhi Biryani	Rice, Chicken, Yogurt, Tomatoes, Potatoes, Sindhi Biryani Masala, Onions	Sindh	Boiling, Layering, I	Boil rice with whole spices. Cook chicken w	Rice
10	Mutton Korma Pulao	Rice, Mutton, Yogurt, Korma Masala, Whole Spices, Onions, Green Chilies	Punjab	Boiling, Simmering	Cook mutton with korma masala, onions, a	Rice
11	Saag Chawal	Rice, Spinach, Mustard Greens, Green Chilies, Garlic, Ghee, Salt	Punjab	Boiling, Sautéing	Boil rice separately. Sauté spinach and mus	Rice
12	Keema Biryani	Rice, Minced Meat, Onions, Tomatoes, Biryani Masala, Yogurt, Green Chi	Punjab	Boiling, Layering, I	Cook minced meat with onions, tomatoes,	Rice
13	Machli Pulao	Rice, Fish, Yogurt, Tomatoes, Onions, Green Chilies, Whole Spices, Ginger	Sindh	Boiling, Simmering	Cook fish with spices and tomatoes. Boil ric	Rice
14	Chicken Qorma Pulao	Rice, Chicken, Yogurt, Qorma Masala, Onions, Ginger, Garlic, Green Chilie	Punjab	Boiling, Simmering	Cook chicken with qorma masala, onions, a	Rice
15	Chicken Shashlik Biryani	Rice, Grilled Chicken, Shashlik Sauce, Onions, Tomatoes, Green Capsicum	Punjab	Grilling, Layering, I	Grill chicken with shashlik sauce. Layer boil	Rice
16	Gosht Palak Chawal	Rice, Mutton, Spinach, Mustard Greens, Green Chilies, Garlic, Ghee	Punjab	Boiling, Simmering	Cook mutton with spinach and mustard gre	Rice
17	Aloo Keema Pulao	Rice, Minced Meat, Potatoes, Onions, Tomatoes, Whole Spices, Ginger, G	Punjab	Boiling, Simmering	Cook minced meat with potatoes, onions, a	Rice
18	Kachhi Biryani	Rice, Raw Marinated Meat, Potatoes, Biryani Masala, Yogurt, Onions, Gin	Sindh	Boiling, Layering, I	Marinate meat with spices and yogurt. Lay	Rice
19	Tandoori Chicken Biryani	Rice, Tandoori Chicken, Yogurt, Biryani Masala, Onions, Ginger, Garlic, Co	Punjab	Boiling, Grilling, Di	Grill tandoori chicken. Boil rice separately.	Rice
20	Qeema Khichdi	Rice, Minced Meat, Moong Dal, Salt, Turmeric, Ghee, Cumin, Green Chilie	Sindh	Boiling, Simmering	Boil rice and moong dal with minced meat,	Rice

Fig. 1: Initial recipes from the dataset

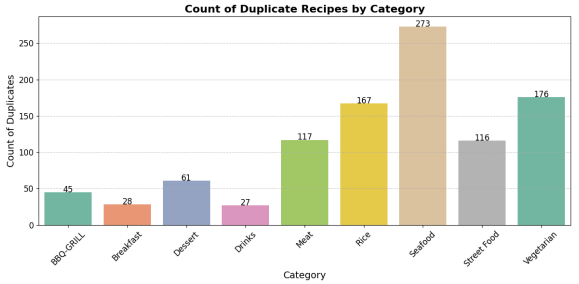


Fig. 2: Count of categories in the dataset

After combining the data from ChatGPT into an excel file, we had to preprocess the data to remove duplicates and combine categories to ensure dataset's quality before sending it to a model. Figure 3 shows a distribution of data in the dataset with duplicates before cleaning.

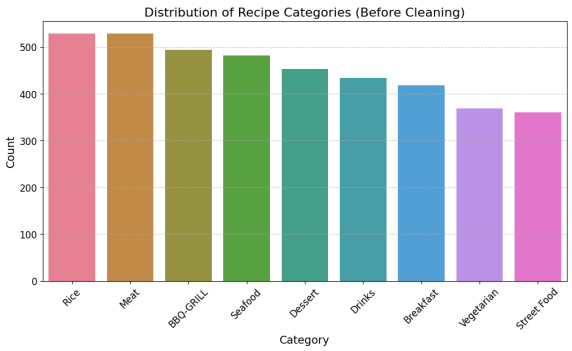


Fig. 3: Dataset categories count before cleaning

To remove duplicate recipes from our dataset, we used a python code and removed the recipes that were redundant in ChatGPT's dataset. Moreover, there were some recipes that were repeated in several categories. To let one recipe belong to more than one category, we implemented a python code that puts repeated recipes in categories as one. Hence, initially we had a data of 4066 and after cleaning we are left with 3036 recipes.

V. Implementation Details

The recipe generator leverages various machine learning models and a cosine similarity-based evaluation system to match user-provided ingredients with corresponding recipes from the dataset. The key components of this system are as follows:

- **Model Selection:** For performance evaluation, we selected several machine learning models including:
  1) **Logistic Regression:** Logistic Regression is a simple yet effective model that works well for problems where the relationship between features (ingredients) and target classes (recipes) can be assumed to be approximately linear. We chose this as our baseline model for evaluation due to its interpretability, combined with its computational efficiency and speed in training, particularly with large datasets. This makes it ideal for initial evaluations in our classification tasks.
  2) **Decision Tree Classifier:** We used Random Forests for our evaluation due to their improved accuracy from aggregating multiple decision trees, resilience to overfitting in imbalanced datasets, and ability to rank ingredient importance, providing insights into key ingredients for specific Pakistani dishes.
  3) **Random Forest Classifier:** We used Random Forests for our evaluation due to their improved accuracy from aggregating multiple decision trees, resilience to overfitting in imbalanced datasets, and ability to rank ingredient importance, providing insights into key ingredients for specific Pakistani dishes.
  4) **Support Vector Machine (SVM)**: We chose Support Vector Machines (SVMs) for our evaluation because they effectively handle high-dimensional data with TF-IDF vectors, excel at modeling non-linear relationships using kernel tricks, and maintain accuracy even with limited datasets, making them ideal for our Pakistani recipe analysis.

  These models were chosen to analyze how effectively they predict recipe names based on input ingredients, measured through cosine similarity between predicted and actual recipes.
- **Performance Metrics:** In our evaluation, we couldn't rely on accuracy metrics due to the nature of our data, where different ingredients can yield similar recipes, such as Biryani and Pulao or Tikka Biryani. These recipes may be classified incorrectly under traditional accuracy measures simply because their names don't match. To address this, we defined our own performance metric using cosine similarity between the ingredient vectors of the predicted recipe and the actual ground truth recipe. We consider a prediction accurate if the cosine similarity score exceeds a defined threshold of 0.7, allowing for a more nuanced evaluation of our models' performance in capturing the relationships between ingredients

and recipes.

- **Key Parameters:** Currently, our parameters consist of a list of ingredients provided by the user. However, our dataset is designed to incorporate additional inputs, such as the category of food (e.g., BBQ, breakfast, drink, dessert) and the region from which the recipe recommendations are needed. The output of our model includes the name of the recipe, its cooking process, the complete recipe, and its category.
- **Software Libraries:** The model is implemented in Python, utilizing key libraries such as:
  1) **Scikit-learn:** For implementing and evaluating machine learning models.
  2) **TfidfVectorizer:** For converting text-based ingredients into numerical vectors for similarity comparison.
  3) **Matplotlib and Seaborn:** For visualizing model performance through confusion matrices and performance graphs.

VI. Evaluation

For our evaluation metrics, we utilized cosine similarity accuracy to measure the effectiveness of the recipe recommendation model. Cosine similarity is a metric that calculates the cosine of the angle between two vectors, which in our case represent the ingredient vectors of the user's input and the recipes in the dataset. It effectively captures the similarity between two sets of ingredients, regardless of their magnitude, focusing on the angle between them. This approach is particularly useful in high-dimensional spaces like ours, where the number of ingredients is large, and we are more interested in the overlap or relationship between ingredients rather than their raw count. By using cosine similarity accuracy, we aim to return recipes that have a high degree of similarity with the user's provided ingredients, ensuring more relevant and accurate recipe suggestions.

We did not use the accuracy metric that is mostly used because for that we require ground truth and actual predictions to calculate it. In our model, the recipes that are being given as an output give a recipe that is close to the given ingredients. There's no exactly correct answer that exists that can be matched.

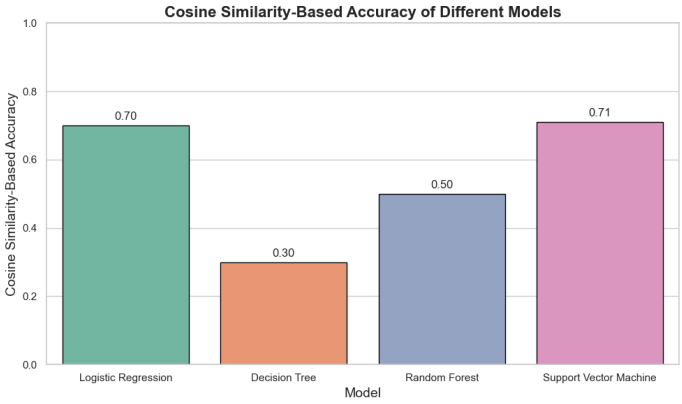


Fig. 4: Cosine-similarity Accuracy of the models

We further did testing on random data to evaluate every models performance. The results of a few test cases are shown below:

```
Predicted Recipes for
"chicken, rice":
LR: Chicken 65 Biryani
DT: Butter Chicken Fries
RF: Chinioti Chicken
SVM: Chicken 65 Biryani

Predicted Recipes for
"vegetables, mutton, garlic, ginger":
LR: Meat Saag Gosht
DT: Chutney Sandwich
RF: Beef Tamatar Gosht
SVM: Meat Saag Gosht

Predicted Recipes for
"milk, sugar, flour, rice":
LR: Bakar Khani
DT: Boondi Phirni
RF: Kheer Pancakes
SVM: Bakar Khani
```

```
Predicted Recipes for
"fish, lemon, herbs":
LR: BBQ Fish & Lemon Herb Rice
DT: Fish and Herb Quinoa
RF: Fish Tikka Wrap
SVM: BBQ Fish & Lemon Herb Rice
```

The model predictions reveal a strong correlation between Linear Regression (LR) and Support Vector Machine (SVM) with the input ingredients, as both models consistently provided the same recipes closely tied to the given ingredients. For example, both RF and SVM accurately predicted "Chicken 65 Biryani" for "chicken, rice" and "BBQ Fish & Lemon Herb Rice" for "fish, lemon, herbs," showing their ability to match recipes with essential components. Although Decision Tree (DT) and Random Forest (RF) also performed well, they occasionally deviated from the input, such as predicting "Butter Chicken Fries" and "Chinioti Chicken" for "chicken, rice," overlooking rice or introducing additional ingredients.

VII. Analysis and Insights

While selecting our baseline model, we observed that the Support Vector Machine (SVM) consistently outperformed the other models in terms of accuracy. This indicates that the SVM model predicts the recipe most closely aligned with the provided ingredients. Although all the models we tested aimed to achieve this, their accuracy varied. The task involved matching the user's input ingredients to the recipe that was most relevant, rather than an exact match. This approach introduces a key limitation: we couldn't calculate accuracy as we would in standard classification models. Since users may omit certain ingredients when providing input, our model returns the recipe that best approximates their query instead of an exact match.

Another limitation is that our current model predicts only a single recipe, but ideally, we want to offer users multiple options. We also noticed that the Logistic Regression model had the fastest runtime compared to the others, while more complex models took longer to compute. The accuracy of the chosen model depends heavily on the dataset split, which in this case was 60% for training and 40% for testing. If we adjust this division, the model's performance is likely to change.

Additionally, we found that the cosine similarity threshold, set at 0.7, produced results closest to our expectations. Adjusting this threshold could significantly impact performance. Finally, traditional classification metrics like F1-score, precision, and recall are not applicable here due to our approach of predicting the closest matching recipe rather than the exact one. Thus, calculating accuracy requires modifications to the algorithm for a more accurate measure of performance.

VIII. Future Work

We plan on adding improvements to our chosen model and also exploring other potential models:

- **Efficiency of SVM:** SVM can become computationally expensive, especially with large datasets. Stochastic Gradient Descent (SGD) is a way to improve the performance. Using SGD for SVM speeds up the training by updating weights based on a small random subset of the data (a "mini-batch") rather than the entire dataset at once. This reduces computational time and resources.
- **Front-end Development:** We plan on creating a simple, intuitive interface where users can input ingredients and see recommendations. We can add filters related to region or type of food and improving visual aspects, such as displaying images of recommended dishes.
- **Generate Multiple Recommendations:** We will modify the SVM model to return the top N recipes based on the highest confidence scores. This would allow users to see a ranked list of suggestions rather than a single result.
- **Cosine-Similarity Accuracy changes:** We can change the cosine-similarity threshold to get a much similar result to the given ingredients. We can also try the Jaccard Index to measure the performance of the models.